07 · Automations & Administration

> *Audience: PMO admins, platform owners · Time: 30 minutes*
> *Prerequisite: modules 01–05 · Screens: Connect group + Administration group*

Learning objectives

Configure the platform for your organisation: roles and permissions, custom fields and form layouts, cross-entity automations, integrations/webhooks — and verify everything through the audit trail.

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1. Control Automations (`#/automations`) — the cross-entity engine

Rules that let entities react to each other automatically. Eleven triggers:

Trigger	Typical use
`TASK_OVERDUE`	Notify owner + escalate to PM after N days
`TIMESHEET_SUBMITTED`	Ping approver; warn if > 7 days unapproved
`TIMESHEET_APPROVED`	Notify finance; update cost alerts
`HEALTH_CHANGED`	Red project → summon PMO + sponsor note
`EVM_BREACH`	CPI/SPI < threshold → corrective-action workflow
`RAID_CREATED`	CRITICAL risk → notify program board
`CHANGE_REQUESTED`	Route CR to approver chain
`MILESTONE_MISSED`	Alert sponsor; flag on dashboards
`SCHEDULE_CHANGED`	Recompute impacts; notify dependents
`STATUS_CHANGED`	Phase transitions → trigger gates/notifications
`MANUAL`	On-demand run from a record

Actions cover notifications, RAID/alert creation, governance updates and webhook calls; **conditions** scope a rule ("only CRITICAL", "only portfolio X") and every execution is logged.

Scan now (`/api/automations/scan`) evaluates rules across the portfolio with 24-hour deduplication — safe to run repeatedly, no duplicate storms.

2. Integration Hub (`#/integrations`)

Connected systems (email, chat, ticketing, ERP connectors): configure credentials, sync direction and health. The CTO Tower (`#/technology`) monitors integration/technical health.

3. Webhooks (`#/webhooks`)

Event subscriptions for your own systems: register an endpoint, choose event types (project updated, alert created, gate decided…), inspect delivery history with retry status. This is how the Control Tower feeds external data warehouses or chatops.

4. AI Connector Gateway (`#/ai-gateway`) & Extension Hub (`#/extensions`)

- **AI Gateway** governs which AI connectors may run, with keys, limits and audit — AI actions stay supervised.
- **Extension Hub** manages optional platform extensions.

5. Administration (`Administration` group)

Screen	Purpose
Users (`#/admin/users`)	Create/disable users, assign roles, reset passwords
Roles & Permissions (`#/admin/roles`)	Role templates + custom roles; permission matrix editor
PMO Template Library (`#/admin/templates`)	Organisational templates (WBS skeletons, report templates, gate criteria)
Audit Trail (`#/admin/audit`)	Immutable log: who changed what, when, before/after
Control Center (`#/admin-center`)	Platform health & config overview
Configuration (`#/admin/settings`)	Org settings, calendars, thresholds, RAG rules
Field Dictionary (`#/admin/fields`)	Dynamic custom fields per entity
Form Tab Builder (`#/admin/tabs`)	Arrange form tabs per entity type

6. Roles — the eight templates

EXECUTIVE (100) · PMO_ADMIN (90, `*`) · PORTFOLIO_MANAGER (80) · PROGRAM_MANAGER (70) · PROJECT_MANAGER (60) · FINANCE (60) · AUDITOR (50, read-only + audit) · TEAM_MEMBER (20).

Each carries a permission set (e.g. `executive.view`, `evm.manage`, `timesheet.approve`, `gate.decide`). Custom roles mix permissions from the matrix. Super admins (`*`) see everything.

> *Teaching point: permissions gate the **menu**, the **UI actions** and the **API** — three layers, same source of truth.*

7. Field Dictionary & Form Tab Builder — adapt without code

1. **Field Dictionary**: add a custom field to any entity — e.g. "Client Region" (select: EMEA/APAC/AMER) on Projects, or "Carbon Estimate" (number) on

Tasks. Fields become filterable, reportable and API-visible instantly.

2. **Form Tab Builder**: reorganise entity forms into tabs that match your process — e.g. Project form → *General* | *Delivery* | *Commercial* | *Custom*.
3. No deployment needed — configuration is live for all users immediately.

8. Audit Trail — the accountability layer

Every create/update/delete and governance decision is recorded with actor, timestamp and before/after snapshots. Filter by entity, user or action. During incidents or audits, this screen answers "who changed this and when" in seconds.

9. Exercise (15 min) — configure like a PMO admin

1. Log in as `pmo@pmct.io`.
2. **Automations** → create: trigger `EVM_BREACH`, condition CPI < 0.90, action → notify finance + create alert. Run **Scan now** — check the execution log.
3. **Roles** → inspect the TEAM_MEMBER template — which permissions does it hold? Which screens disappear for Liam because of that?
4. **Field Dictionary** → add "Client Region" (select) to Projects.
5. **Form Tab Builder** → move it onto a "Commercial" tab of the Project form. Open a project — verify the field is live.
6. **Audit Trail** → find your own changes from this session.

Discussion: step 4–5 extended the data model and UX in under a minute with no code and no deployment. What would your organisation configure first?

10. Key takeaways

- Automations connect entities: one event can cascade across the portfolio, deduplicated and audited.
- Roles = least privilege; Field Dictionary + Tab Builder = adapt without code.
- The Audit Trail makes the whole platform accountable by default.